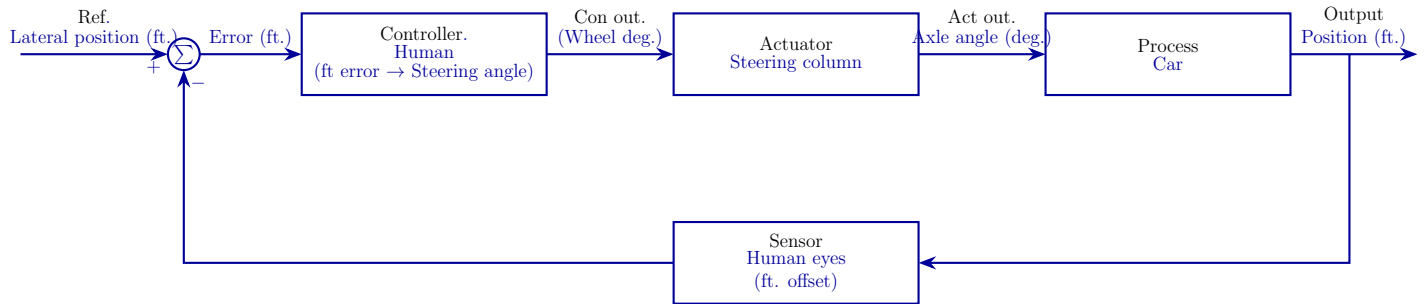


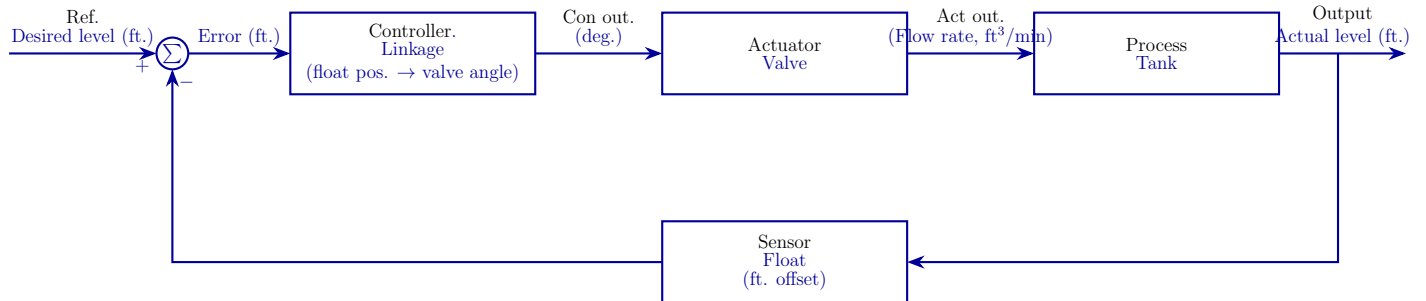
Problem 1 — 10 points

Draw a component block diagram for each of the following feedback control systems:

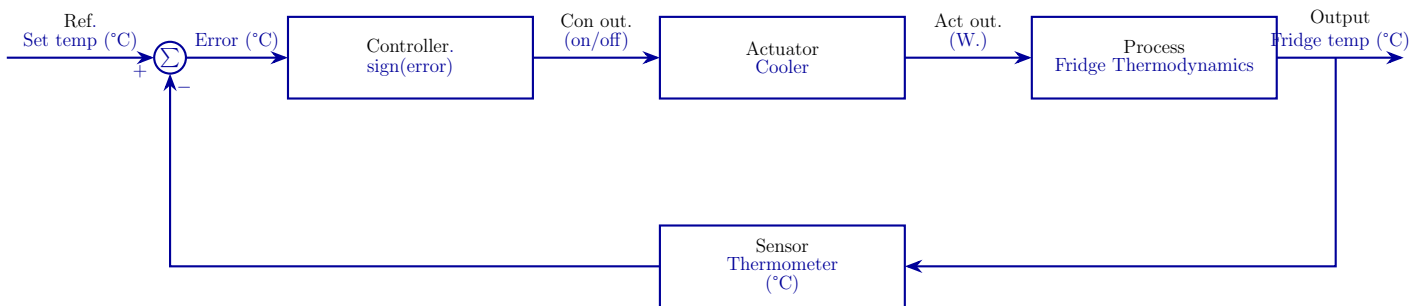
a. The manual steering system of an automobile



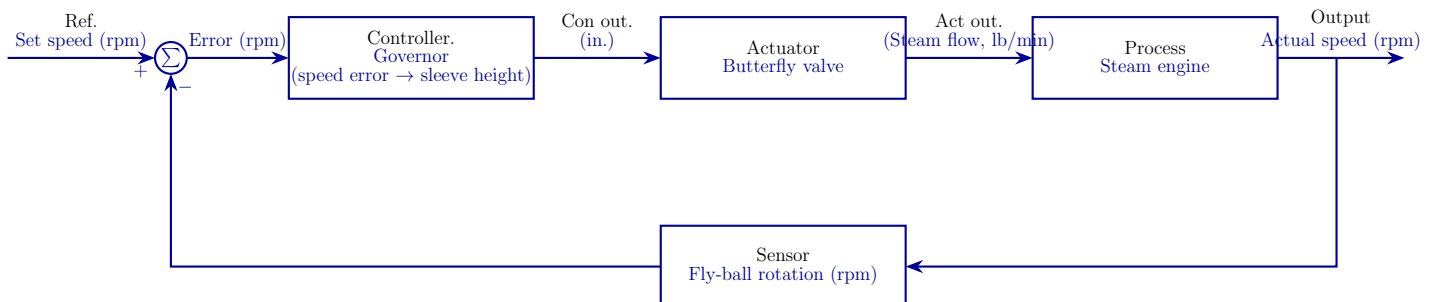
b. The water level controlled by a float and valve

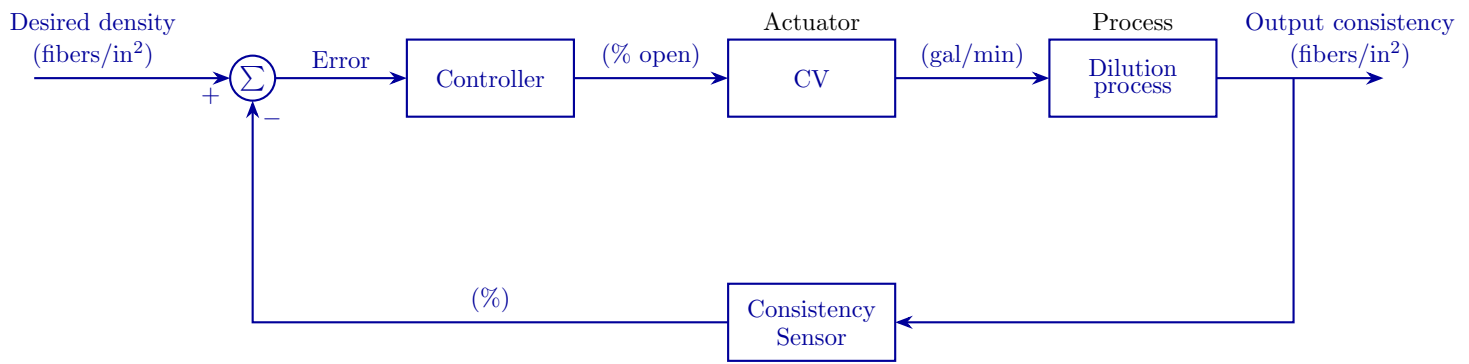


c. Temperature control in a refrigerator

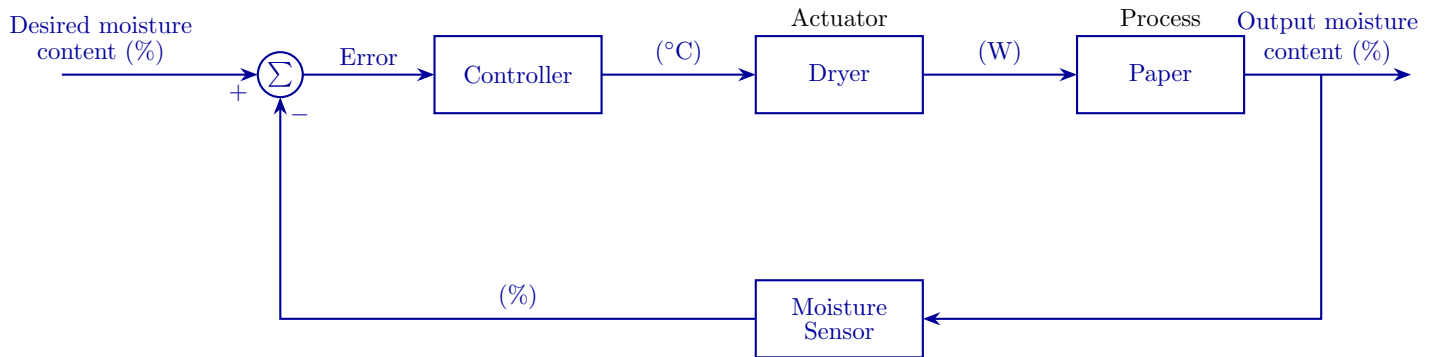


d. Watt's steam engine with fly-ball governor





b. Control of moisture



Problem 3 — 10 points

Find the Laplace transform for the following time functions:

a. $f(t) = \sin(3t) + 2 \cos(3t) + e^{-t} \sin(3t)$

$$\mathcal{L}[f] = \frac{3}{s^2 + 9} + \frac{2s}{s^2 + 9} + \frac{3}{(s + 1)^2 + 9}$$

b. $f(t) = t \sin t$

$$\mathcal{L}[f] = \frac{2s}{(s^2 + 1)^2}$$

c. $f(t) = 4 + 7t + t^2 + \delta(t)$

$$\mathcal{L}[f] = \frac{4}{s} + \frac{7}{s^2} + \frac{2}{s^3} + 1$$

Problem 4 — 10 points

Using the convolution integral, find the step response of the system whose impulse response is given below:

$$h(t) = \begin{cases} te^{-t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Verify the results using the Laplace transform.

$$\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} u(\tau)h(t-\tau)d\tau \\
&= \int_0^t (t-\tau)e^{-(t-\tau)}d\tau \\
&= e^{-t} \int_0^t (t-\tau)e^{\tau}d\tau \\
&= e^{-t} \left[(t-\tau)e^{\tau} \Big|_0^t + \int_0^t e^{\tau}d\tau \right] \\
&= e^{-t} [0-t+(e^t-1)] \\
&= e^{-t}(e^t-1-t) \\
&= \boxed{1-e^{-t}-te^{-t}}
\end{aligned}$$

Or, via Laplace:

$$\mathcal{L}[h(t)] = H(s) = \frac{1}{(s+1)^2}$$

so

$$Y(s) = H(s) \cdot U(s) = \frac{1}{(s+1)^2} \cdot \frac{1}{s} = \frac{1}{s(s+1)^2}$$

which is exactly

$$\mathcal{L}[1-e^{-t}-te^{-t}] = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} = \frac{1}{s(s+1)^2}$$

Problem 5 — 10 points

Solve the following ODEs using the Laplace transform:

a. $\ddot{y}(t) + y(t) = t; \quad y(0) = 1; \quad \dot{y}(0) = -1$

$$\begin{aligned}
\mathcal{L}[\ddot{y} + y] &= \mathcal{L}[t] \\
s^2Y(s) - sy(0) - \dot{y}(0) + Y(s) &= \frac{1}{s^2} \\
(s^2 + 1)Y(s) - s + 1 &= \frac{1}{s^2}
\end{aligned}$$

$$\begin{aligned}
Y(s) &= \frac{1}{s^2(s^2+1)} + \frac{s-1}{s^2+1} \\
&= \frac{1}{s^2} - \frac{1}{s^2+1} + \frac{s}{s^2+1} - \frac{1}{s^2+1} \\
y(t) &= t - \sin t + \cos t - \sin t \\
&= \boxed{t + \cos t - 2\sin t}
\end{aligned}$$

b. $\ddot{y}(t) + 2\dot{y}(t) = e^t; \quad y(0) = 1; \quad \dot{y}(0) = 2$

$$\begin{aligned}
(s^2 Y(s) - s y(0) - \dot{y}(0)) + 2(s Y(s) - y(0)) &= \frac{1}{s-1} \\
(s^2 + 2s)Y(s) - (s+2)y(0) - \dot{y}(0) &= \frac{1}{s-1} \\
s(s+2)Y(s) - (s+2) - 2 &= \frac{1}{s-1} \\
Y(s) &= \frac{1}{(s-1)s(s+2)} + \frac{s+4}{s(s+2)} \\
&= -\frac{1}{2s} + \frac{1}{6(s+2)} + \frac{1}{3(s-1)} + \frac{2}{s} - \frac{1}{s+2} \\
y(t) &= -\frac{1}{2} + \frac{1}{6}e^{-2t} + \frac{1}{3}e^t + 2 - e^{-2t} \\
&= \boxed{-\frac{5}{6}e^{-2t} + \frac{1}{3}e^t + \frac{3}{2}}
\end{aligned}$$

Problem 6 — 10 points

Find the time function corresponding to each of the following Laplace transforms:

a. $F(s) = \frac{3s^2 + 6s + 6}{(s+1)(s^2 + 6s + 10)}$

$$\begin{aligned}
F(s) &= \frac{3s^2 + 6s + 6}{(s+1)(s^2 + 6s + 10)} \\
&= \frac{12s}{5(s^2 + 6s + 10)} + \frac{3}{5(s+1)} \\
&= \frac{12}{5} \frac{s}{(s+3)^2 + 1} + \frac{3}{5} \frac{1}{s+1} \\
&= \frac{12}{5} \left[\frac{s+3}{(s+3)^2 + 1} - \frac{3}{(s+3)^2 + 1} \right] + \frac{3}{5} \frac{1}{s+1} \\
f(t) &= \frac{12}{5} [e^{-3t} \cos t - 3e^{-3t} \sin t] + \frac{3}{5} e^{-t} \\
&= \boxed{\frac{12}{5} e^{-3t} (\cos t - 3 \sin t) + \frac{3}{5} e^{-t}}
\end{aligned}$$

b. $F(s) = \frac{e^{-s}}{s^2}$

$$\begin{aligned}
F(s) &= \frac{1}{s^2} e^{-s} \\
f(t) &= \boxed{(t-1)u(t-1)}
\end{aligned}$$

Note: You can use integration tables. Use the MATLAB Symbolic Algebra toolbox for verification.